Predicting Diabetic Status Among Females: A Logistic Regression Analysis of Clinical Indicators and Age-Based Data Gaps

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PROBLEM

Millions of Americans live with Diabetes, a chronic metabolic disease that has created an epidemic in the United States with a higher rate in women. There is not sufficient attention and research in understanding what and how demographic/medical factors play a role in diabetic status specifically in women. Accurate predictions motivate for faster diagnosis and treatment.

BACKGROUND

Individuals with Type I Diabetes are unable to make insulin and individuals with type II Diabetes can make insulin but their cells are unable to use it. Specifically, women are more likely to have family history of diabetes and more at risk due to their body mass index (Gucciardi et al., 2008). In a study by Weinstein et al, body mass index was a key indicator in developing diabetes (2004). Rajendra et al., suggests logistic regression is an efficient model in predicting clinical outcomes with use of additional factors including cross-validation, feature selection and null value removal. To accomplish this, machine learning can be used to create a model built on demographic data and patient medical history to identify new patterns and relationships among health factors that increase and accurately predict Diabetes status in women.

APPROACH

Goals

- Identify key patterns in female patients that have and do not have diabetes
- Identify relationships between medical and demographic features
- Model diabetic status outcome and model accuracy

Data Inspection and Cleaning

The dataset that was used for this analysis was 'diabetes_dataset_prediction_csv' (100,000 records, 9 features) that included demographic and medical information with diabetes status. Predictor variables included gender: female and male, age: 0.08-80, hypertension: yes 1 no 0, heart_disease: yes 1 no 0, smoking_history: current, not current, no info, never, ever, Bmi: body mass: index value, HbA1c levels: measures blood glucose levels in the recent 2 months, Blood glucose level: current blood sugar levels, Diabetes: diabetic 1/not diabetic 0. There was no null values or missing values but duplicated values (3854 records) that were removed at this step. The data was filtered by gender (Female) for this analysis and determined to be unbalanced and was accounted in model.

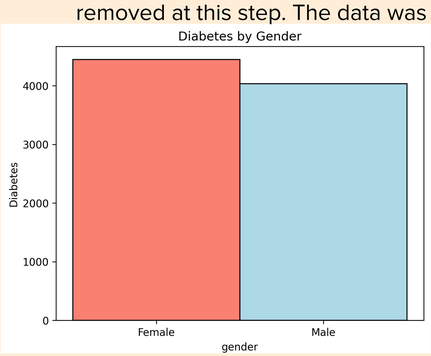


Figure 1 display a histogram of diabetic status of females and males. There were more female with a diabetic status over men in the data set (Female: 56161 records, Male: 39967 records)

Exploratory Data Analysis (EDA)

There are clear indicators of insufficient medical information among different age groups associated with diabetes. Girls under the age of 18 with a diabetes status had minimum medical information compared to other age groups (19-30, 31-45, 46+). Women, 46 and older, had more medical information. Females with a diabetic status, had 4447 records while females without diabetic status, had 51714 records. Specifically girls under 18, had 52 diabetic status records and 8790 undiabetic records. **Figure 2 presents a histogram displaying the differences in records by age group. (18 and under red, 19-30 orange, 31-45 green, 46+ purple).** Blood glucose levels (0.43) and HbA1c levels (0.42) in women both have the highest correlation to diabetes status.

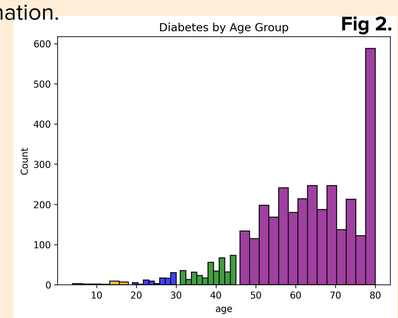


Fig 2.

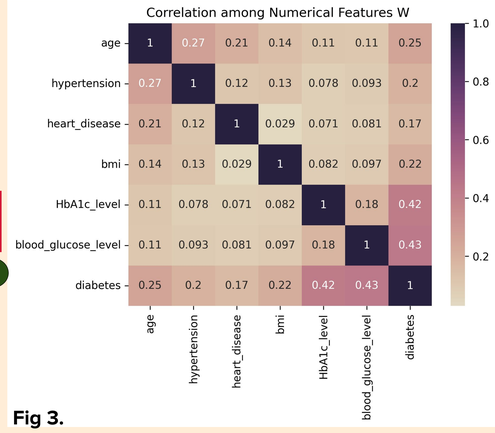


Fig 3.

Figure 3. displays a correlation heat map among numerical features and associated correlations. Blood glucose levels (0.43) and HbA1c levels (0.42) in women both have the highest correlation to diabetes status. Body Mass Index (BMI) holds the second highest correlation (0.24) to diabetic status. **Figure 4. Categorical plot visualizing the relationships between age, diabetes and HbA1c levels.**

Older diabetic women had high HbA1c levels compared to women across all ages that were not diabetic. Fig 4 shows a clear relationship of HbA1C levels regardless of age in diabetic individuals. There is also insufficient information of girls under 18 with HbA1c levels in the data.

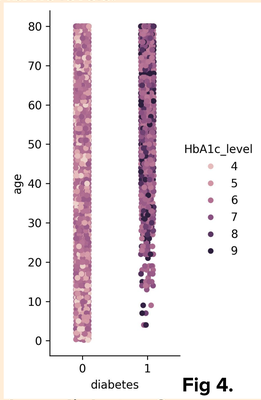


Fig 4.

Logistic Regression Model for diabetes outcome: Diabetes 1, no diabetes 0

To evaluate how these features can predict a condition like diabetic status, a supervised algorithm was used with diabetes as the outcome/response variable and the demographic and medical information as predictor variables. The diabetes outcome variable includes binomial values making logistic regression the best model to use. The feature, smoking_history was engineered from phrases to one word and encoded into numerical values. A train/test split was created with a 80% train and 20% test size, 42 random states, and a stratify parameter to maintain class proportions. The medical/demographic features from the original and test/train data were scaled using StandardScaler(), and the LogisticRegressionCV() model was used with 5-Fold cross-validation, with class_weight="balanced" parameter to handle the unbalanced data. Log-odds equation:

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1(Age) + \beta_2(hypertension) + \beta_3(heart\ disease) + \beta_4(smoking\ history) + \beta_5(bmi) + \beta_6(HbA1c\ level) + \beta_7(blood\ glucose\ level) \quad (1)$$

P: probability of having diabetes, *1-p:* probability of not having diabetes, β: log-odds when all predictors are 0, $\beta_1 \ldots \beta_5$: coefficients for each feature. Below is the log-odds after model:

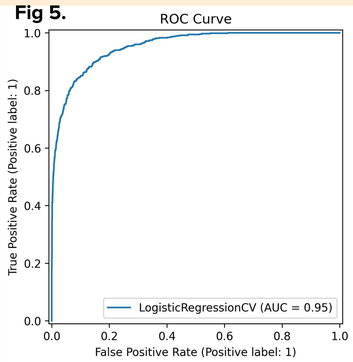
$$\ln\left(\frac{p}{1-p}\right) = -0.6685 + 0.2603\beta_1 + 0.1389\beta_2 + 0.1060\beta_3 - 0.0114\beta_4 + 0.2053\beta_5 + 0.4402\beta_6 + 0.3831\beta_7 \quad (2)$$

EVALUATIONS

TABLE 1.	Precision	recall	fl-score	support
0	0.98	0.90	0.94	6494
1	0.47	0.85	0.60	698
accuracy			**0.89**	7192
Macro avg	0.73	0.87	0.77	7192
Weighted avg	0.93	0.89	0.91	7192

There was a significant difference in data regarding diabetic status for different age groups. There was more data for adult women over 45 years old in comparison to young girl under 18 years of age creating a skewed left-skewed distribution. HbA1c and blood glucose levels are key indicators for diabetic status. They demonstrate the higest correlation with high HbA1c levels consistnt with diabetic individuals. The ROC in **Fig 5** curve illustrates a high true positive rate with a relatively low false positive rate with a AUC=0.95. The model correctly identified 5,821 non-diabetic individuals (TN), accurately predicted 594 diabetic cases (TP), 104 instances where diabetic status was missed (FN), and incorrectly labeled 673 individuals as diabetic (FP). Classification summary includes a 89% accuracy on imbalanced data.

Fig 5.



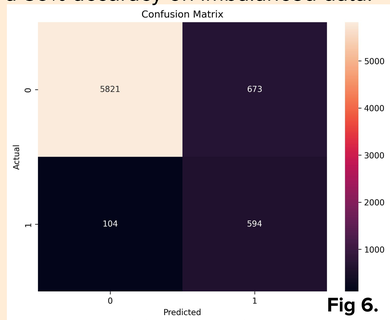


Fig 6.

Figure 5. Receiving Operating Characteristic (ROC) curve was used to plot the true positivity rate against the the False positivity rate. Figure 6. Confusion Matrix was used to compare predicted outcomes against actual outcomes.

CONCLUSIONS

- Age-related data gaps supports the need for more research on younger adults and diabetes risk.
- A clear gender-disparity in data.
- Key medical indicators for diabetic status includes high HbA1c and blood glucose levels across all age groups along with BMI.
- Imbalanced data created irregular prediction, in-depth data inspection required.
- Future Implications: Accurate predictive models can be a great resource for disease prediction, but more relevant research can help fill in the criitcal gap for disease prediction.

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